Supplementary Material: Reference Requirements of the Benchmark

Anonymous Author(s)

Anonymous Institution

This table lists every row and atom of the five reference sets (version 3). Rows are the published reference rows and rows added from the same reference document (marked †); each row is followed by its atoms, the scoring units. Row statements are curator paraphrases. Atom statements are written in one standard form: a content atom says what users can know, find, access, cite or use (*Users can ...*), a profile atom what the profile does (*The profile ...*), and a system atom names the system or role the source names. The reference documents' own wording is cited in the benchmark files. A provisional adjudication of reviewed items was applied on 23 October 2026: one duplicate merged, one umbrella atom left out of scoring, two compound atoms split, three evidence labels downgraded and one omitted need restored; each change is logged in the benchmark files.

Columns. Type: content (c), information a catalog record should carry; profile (p), how the profile is specified (principles, scope, reuse and alignment, obligation levels, code lists, documentation); or system (s), the systems and processes around the profile (governance and maintenance, validation, interfaces, exchange, review)[citation needed]. Only content atoms are scored. *Status* in the source: requirement (req.), proposal (prop.), open issue (open), guidance (guid.) or deferred to future work (def.). *Base*: relation to the pinned DCAT-AP release: restates, tightens, extends, partly covered by, or beyond the base; – for atoms that are not about metadata terms, ? if unresolved. *Raw* and *Cons.*: whether the raw corpus and, for mobilityDCAT-AP and HealthDCAT-AP, the pre-consolidation corpus state the atom: S stated, P in part, U uncertain, N not stated, – no such corpus. For the 196 scored atoms, Raw shows the curator-adjudicated label after two AI reviews (3 Nov 2026; one review was not blind). Cons. uses that adjudication only where consolidation was explicitly reviewed; otherwise it retains the original AI label. All 124 unscored atoms retain their original AI labels (22 Oct 2026). This is not independent human source validation; see evidence-review/adjudication-2026-11-03.json. *Sc.*: scored by default (✓); m merged into an equivalent atom of another row (`Req-24.1` into `Req-19.7`); u an umbrella of separately scored atoms (`H-4.1` of `H-16.1`, `H-26.1`, `H-28.1`, `H-36.1`, `H-66.2`).

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
mobilityDCAT-AP NAPCORE requirement analysis (2022), Table 1; base DCAT-AP 2.0.1; inspected during development: outputs 40 rows (40 published), 98 atoms, 39 scored (39 requirements), 7 restate the base, 30 stated (39 at least in part); pre-consolidation corpus: 38 (39)							
`Req-1` Use sufficiently expressive, context-aware and modular descriptions with explicit structure, types and rules; distinguish entity identities, preserve references, and support persistent identifiers. *(Table 1, PDF pp. 15–19)*							
`Req-1.1`	The profile makes metadata descriptions expressive.	p	req.	–	S	S	
`Req-1.2`	The profile makes metadata descriptions rich.	p	req.	–	S	U	
`Req-1.3`	The profile describes the data adequately.	p	req.	–	S	S	
`Req-1.4`	The profile supports easy access to the data.	p	req.	–	S	S	
`Req-1.5`	The profile makes metadata descriptions context-sensitive.	p	req.	–	U	S	
`Req-1.6`	The profile makes metadata descriptions modular.	p	req.	–	S	S	
`Req-1.7`	The profile gives metadata descriptions a detailed structure.	p	req.	–	S	S	
`Req-1.8`	The profile uses strong typing and rules in metadata descriptions.	p	req.	–	S	S	
`Req-1.9`	The profile keeps referential integrity.	p	req.	–	S	U	
`Req-1.10`	The profile enables persistent identifiers.	p	req.	–	S	S	
`Req-2` Make data findable and reusable; provide a basis for NAP metadata databases, describe catalogues, data and their delivery options, and support metadata federation. *(Table 1, PDF pp. 15–19)*							
`Req-2.1`	The profile makes datasets discoverable and searchable.	p	req.	–	S	S	
`Req-2.2`	The profile makes datasets reusable.	p	req.	–	S	S	
`Req-2.3`	The profile serves as a baseline for developing a NAP metadata database.	p	req.	–	S	S	
`Req-2.4`	The profile allows a detailed description of the registry (catalogue).	p	req.	–	S	S	
`Req-2.5`	The profile allows a detailed description of the registry's datasets.	p	req.	–	S	S	
`Req-2.6`	The profile allows a detailed description of the datasets' distributions and/or data services.	p	req.	–	S	S	
`Req-2.7`	The profile allows aggregation and exchange of metadata.	p	req.	–	S	S	
`Req-3` Expose descriptions in forms suitable for people and software. *(Table 1, PDF pp. 15–19)*							
`Req-3.1`	The profile provides for metadata in a human-readable format.	p	req.	–	S	S	
`Req-3.2`	The profile provides for metadata in a machine-readable format.	p	req.	–	S	S	
`Req-4` Accommodate descriptions in several languages. *(Table 1, PDF pp. 15–19)*							
`Req-4.1`	The profile provides for metadata information in multiple languages.	p	req.	–	S	S	
`Req-5` Use shared vocabularies so participants interpret metadata consistently. *(Table 1, PDF pp. 15–19)*							
`Req-5.1`	The profile adopts a standardised set of vocabularies to ensure semantic interoperability.	p	req.	–	S	S	
`Req-6` Use a common serialization that software agents can process automatically. *(Table 1, PDF pp. 15–19)*							
`Req-6.1`	The profile adopts a standard format or serialisation, for technological interoperability and automated access by software agents.	p	req.	–	S	S	
`Req-7` Publish freely accessible documentation on the Web and, where feasible, provide digitally annotated repository documentation. *(Table 1, PDF pp. 15–19)*							
`Req-7.1`	The profile's documentation is publicly available on the Web without proprietary licences.	s	req.	–	P	S	

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
Req-7.2	The profile's documentation is annotated digitally in a public repository, where possible.	s	req.	–	P	P	
Req-8 Establish governance and revision procedures, public issue discussion and release planning, and incorporate resolved changes in later versions. *(Table 1, PDF pp. 15–19)*							
Req-8.1	The profile defines the governance processes for its maintenance.	s	req.	–	S	S	
Req-8.2	The profile defines its revision methodology.	s	req.	–	S	S	
Req-8.3	The profile allows public commenting on its development.	s	req.	–	S	S	
Req-8.4	The profile has an issues and releases planning tool.	s	req.	–	S	S	
Req-8.5	The profile's authors discuss and resolve issues and proposals and incorporate them into follow-up releases.	s	req.	–	S	S	
Req-9 Build on DCAT-AP terms when suitable and follow its extension rules for additions. *(Table 1, PDF pp. 15–19)*							
Req-9.1	The profile reuses DCAT-AP classes and properties where possible.	p	req.	–	S	S	
Req-9.2	The profile adds any further classes and properties according to the DCAT-AP extension rules.	p	req.	–	S	S	
Req-10 Account for subsequent changes to the DCAT-AP baseline. *(Table 1, PDF pp. 15–19)*							
Req-10.1	The profile follows the ongoing evolution of DCAT-AP, considering its developments and revisions.	p	req.	–	S	S	
Req-11 Review the complete CMC element set against mobility catalogue needs. *(Table 1, PDF pp. 15–19)*							
Req-11.1	The profile considers all CMC metadata elements, to identify metadata that better fit the specific requirements of NAPs.	p	req.	–	S	S	
Req-12 Account for other DCAT-AP profiles and INSPIRE metadata practice. *(Table 1, PDF pp. 15–19)*							
Req-12.1	The profile considers existing DCAT-AP extensions (e.g. GeoDCAT-AP for geospatial data).	p	req.	–	S	S	
Req-12.2	The profile considers the metadata specifications of the INSPIRE Directive.	p	req.	–	S	S	
Req-13 Preserve interoperability with code lists already deployed by NAP operators. *(Table 1, PDF pp. 15–19)*							
Req-13.1	The profile takes into account the vocabularies already used in operational NAPs when defining controlled vocabularies, for interoperability with existing solutions.	p	req.	–	P	S	
Req-14 Describe the varied mobility domain in enough detail to distinguish its growing dataset inventory. *(Table 1, PDF pp. 15–19)*							
Req-14.1	The profile provides exhaustive metadata descriptions for mobility data, given the growing number and variety of mobility datasets.	p	req.	–	P	S	
Req-15 Define a common essential element set while permitting deployment-specific additions. *(Table 1, PDF pp. 15–19)*							
Req-15.1	The profile proposes essential metadata elements relevant to any NAP, as a minimum set of metadata for European NAPs.	p	req.	–	S	S	
Req-15.2	The profile allows further metadata elements in individual NAP deployments.	p	req.	–	S	S	
Req-16 Examine the established CMC/DCAT-AP correspondences to identify unrepresented information. *(Table 1, PDF pp. 15–19)*							
Req-16.1	The profile uses the canonical mappings between DCAT-AP and the CMC to identify information that DCAT-AP cannot yet represent.	p	req.	–	P	S	

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
Req-17 Record creation and revision times, description language, and responsibility for maintaining a metadata record. *(Table 1, PDF pp. 15–19)*							
Req-17.1	Users can know the date and time a metadata record was created or last modified.	c	req.	restates	S	S	✓
Req-17.2	Users can know the language in which the metadata are written.	c	req.	tightens	S	S	✓
Req-17.3	Users can know who is responsible for creating and maintaining the metadata.	c	req.	extends	S	S	✓
Req-18 State how the metadata record itself may be reused. *(Table 1, PDF pp. 15–19)*							
Req-18.1	Users can know the conditions of use of the metadata itself.	c	req.	restates	S	S	✓
Req-19 Characterise the data using a title and synopsis, resource kind, dataset and service classifications including MMTIS service categories, language and payload georeferencing. *(Table 1, PDF pp. 15–19)*							
Req-19.1	Users can know the name of a resource.	c	req.	restates	S	S	✓
Req-19.2	Users can know a brief description of a resource.	c	req.	restates	S	S	✓
Req-19.3	Users can know the type of a resource (dataset or service).	c	req.	restates	S	S	✓
Req-19.4	Users can know the dataset type category, by logical clustering or by the EC Delegated Regulations.	c	req.	beyond	S	S	✓
Req-19.5	Users can know the service type category listed in Delegated Regulation (EU) 2017/1926.	c	req.	beyond	P	P	✓
Req-19.6	Users can know the language of a resource.	c	req.	restates	S	S	✓
Req-19.7	Users can know the georeferencing method applied within the payload.	c	req.	beyond	P	S	✓
Req-20 Indicate which ITS delegated regulations apply to a resource. *(Table 1, PDF pp. 15–19)*							
Req-20.1	Users can know which Delegated Regulation (MMTIS, RTTI, SRTI, SSTP) covers a dataset or service.	c	req.	beyond	P	S	✓
Req-21 Coordinate, host, maintain and extend multilingual code lists for describing dataset and service content. *(Table 1, PDF pp. 15–19)*							
Req-21.1	The profile supports the coordinated development of a controlled vocabulary for describing dataset and service content.	p	req.	–	S	S	
Req-21.2	The profile hosts the controlled vocabulary.	p	req.	–	P	S	
Req-21.3	The profile keeps the controlled vocabulary maintained.	p	req.	–	S	S	
Req-21.4	The profile keeps the controlled vocabulary extendable.	p	req.	–	S	S	
Req-21.5	The profile allows the controlled vocabulary to be translated into local languages, for harmonisation.	p	req.	–	S	S	
Req-22 Represent the interval over which the published data remain valid. *(Table 1, PDF pp. 15–19)*							
Req-22.1	Users can know the period in which the published data do not expire.	c	req.	extends	S	S	✓
Req-23 Describe spatial extent, applicable regions and network coverage using established functional classifications. *(Table 1, PDF pp. 15–19)*							
Req-23.1	Users can know the area a dataset covers.	c	req.	tightens	S	S	✓
Req-23.2	Users can know the regions in which a dataset is valid.	c	req.	tightens	S	S	✓
Req-23.3	Users can know the transport network a dataset considers, following existing, widely applicable functional classifications.	c	req.	beyond	P	S	✓
Req-24 Identify payload location-reference methods and coordinate reference systems. *(Table 1, PDF pp. 15–19)*							
Req-24.1	Users can know the location referencing methods used in a dataset or service.	c	req.	beyond	S	S	m
Req-24.2	Users can know the coordinate reference systems used in a dataset or service.	c	req.	tightens	S	S	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
Req-25 Identify the transport modes and operators represented by the data. *(Table 1, PDF pp. 15–19)*							
Req-25.1	Users can know the transport modes a dataset deals with.	c	req.	beyond	S	S	✓
Req-25.2	Users can know the transport operators a dataset deals with.	c	req.	beyond	S	S	✓
Req-26 Identify the publishing and owning parties for a dataset. *(Table 1, PDF pp. 15–19)*							
Req-26.1	Users can know who publishes a dataset.	c	req.	tightens	S	S	✓
Req-26.2	Users can know who owns a dataset.	c	req.	beyond	S	S	✓
Req-27 Describe contractual, licensing and other conditions on data use. *(Table 1, PDF pp. 15–19)*							
Req-27.1	Users can know whether a licence governs the use of the data.	c	req.	tightens	S	S	✓
Req-27.2	Users can know whether a contract governs the use of the data.	c	req.	extends	S	S	✓
Req-27.3	Users can know any other condition for using the data.	c	req.	extends	S	S	✓
Req-28 Describe the encoding, syntax, grammar and underlying data model used for access. *(Table 1, PDF pp. 15–19)*							
Req-28.1	Users can know the encoding of the data for access.	c	req.	beyond	S	S	✓
Req-28.2	Users can know the syntax of the data.	c	req.	tightens	S	S	✓
Req-28.3	Users can know the grammar of the data.	c	req.	beyond	S	S	✓
Req-28.4	Users can know the data model of the data.	c	req.	beyond	S	S	✓
Req-29 Offer schema information and example data so consumers can assess suitability before integration. *(Table 1, PDF pp. 15–19)*							
Req-29.1	Users can know the data schema, so they can evaluate the data before setting up consumption.	c	req.	beyond	S	S	✓
Req-29.2	Users can access data samples, so they can evaluate the data before setting up consumption.	c	req.	extends	S	S	✓
Req-30 Distinguish precise data-model versions, publication types and extensions, such as those used in DATEX II, for discovery. *(Table 1, PDF pp. 15–19)*							
Req-30.1	Users can know precisely the version of the data model a dataset adopts (e.g. the DATEX II version), to allow better searching.	c	req.	beyond	P	S	✓
Req-30.2	Users can know the publication type of the data model a dataset adopts (e.g. the DATEX II publication type), to allow better searching.	c	req.	beyond	P	S	✓
Req-30.3	Users can know the extensions of the data model a dataset adopts (e.g. DATEX II extensions), to allow better searching.	c	req.	beyond	P	S	✓
Req-31 Record update cadence, quality criteria and the history and status of ITS-regulation compliance assessments. *(Table 1, PDF pp. 15–19)*							
Req-31.1	Users can know the update rate of a dataset.	c	req.	tightens	S	S	✓
Req-31.2	Users can know the quality criteria of a dataset.	c	req.	beyond	S	S	✓
Req-31.3	Users can know the history of the procedures that assess compliance with the Delegated Regulations on providing data via a NAP.	c	req.	beyond	P	S	✓
Req-31.4	Users can know the status of the procedures that assess compliance with the Delegated Regulations on providing data via a NAP.	c	req.	beyond	S	S	✓
Req-32 Define occurrence limits and obligation levels and explain their consequences for database storage, user interfaces and harvesting, consistently with DCAT-AP. *(Table 1, PDF pp. 15–19)*							
Req-32.1	The profile specifies the cardinality of each metadata element.	p	req.	–	S	S	
Req-32.2	The profile specifies the obligation of each metadata element (mandatory, recommended, optional).	p	req.	–	S	S	

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
Req-32.3	The profile establishes clear rules on the cardinalities of its metadata elements for NAP deployments (e.g. their implications for database set-ups, GUI designs and API or harvesting mechanisms).	p	req.	–	P	S	
Req-32.4	The profile considers the existing DCAT-AP rules on cardinalities.	p	req.	–	S	S	
Req-33	Provide identifier guidance. *(Table 1, PDF pp. 15–19)*						
Req-33.1	The profile defines guidelines for identifiers.	p	req.	–	S	S	
Req-34	Define permitted values for metadata elements, using suitable controlled vocabularies. *(Table 1, PDF pp. 15–19)*						
Req-34.1	The profile defines the expected value range of each metadata element (for example with controlled vocabularies).	p	req.	–	S	S	
Req-35	Track revisions and versions of catalogue metadata. *(Table 1, PDF pp. 15–19)*						
Req-35.1	Users can know the changes of metadata records on a NAP.	c	req.	restates	S	S	✓
Req-35.2	Users can know the different versions of metadata records on a NAP.	c	req.	extends	P	S	✓
Req-36	Enable automated conformance checking of metadata. *(Table 1, PDF pp. 15–19)*						
Req-36.1	The profile supports the validation, possibly automatic, of metadata that adopts it.	s	req.	–	S	S	
Req-37	Explain element meanings and how to apply them within NAP operations. *(Table 1, PDF pp. 15–19)*						
Req-37.1	The profile provides usage rules in addition to the metadata definitions, explaining the semantics of each metadata element.	p	req.	–	S	S	
Req-37.2	The profile advises how to relate the metadata elements to NAP processes.	p	req.	–	P	S	
Req-38	Explain allocation of metadata between a dataset or service and its multiple delivery resources. *(Table 1, PDF pp. 15–19)*						
Req-38.1	The profile provides guidelines and solutions for dividing metadata between a dataset or service and its multiple resources.	p	req.	–	S	S	
Req-39	Explain exchange of mobility-specific descriptions with other-domain portals, including handling extensions during harvesting. *(Table 1, PDF pp. 15–19)*						
Req-39.1	The profile provides guidelines on making NAP-specific metadata interoperable with other data portals (e.g. outside the mobility domain).	s	req.	–	S	S	
Req-39.2	The profile's interoperability guidelines cover how napDCAT-AP extensions are handled when harvesting from other portals.	s	req.	–	P	S	
Req-40	Give guidance for assuring the quality of metadata received by a NAP. *(Table 1, PDF pp. 15–19)*						
Req-40.1	The profile provides guidelines for a quality assurance process that checks the quality of metadata collected on a NAP.	s	req.	–	S	S	

GeoDCAT-AP GeoDCAT-AP 1.0 (2015); base DCAT-AP 1.1; inspected during development: no

46 rows (7 published), 56 atoms, 37 scored (37 requirements), 16 restate the base, 37 stated (37 at least in part)

G-1	Provide RDF representations of spatial metadata that conform to DCAT-AP, retain meaning as far as practical and support cross-domain reuse. *(GeoDCAT-AP 1.0 ğ1.1;4.1)*						
G-1.1	The profile provides a DCAT-AP-conformant representation of geospatial metadata.	p	req.	–	P	–	
G-1.2	The profile provides an RDF representation of geospatial metadata that is as comprehensive as possible and avoids semantic loss.	p	req.	–	S	–	
G-1.3	The profile promotes the cross-domain reuse of geospatial metadata.	p	req.	–	P	–	

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
G-2 Include every mandatory baseline DCAT-AP element. *(GeoDCAT-AP 1.0 ğ4.1)*							
G-2.1	The profile includes, at least, all mandatory DCAT-AP elements.	p	req.	–	U	–	
G-3 Use other vocabularies for geospatial information only where the baseline does not cover it. *(GeoDCAT-AP 1.0 ğ4.1)*							
G-3.1	The profile uses vocabularies other than DCAT-AP only for geospatial metadata elements that DCAT-AP does not support.	p	req.	–	P	–	
G-4 Base alignments on established practice and working-group implementation experience. *(GeoDCAT-AP 1.0 ğ4.1)*							
G-4.1	The profile bases its alignments as far as possible on existing practice, in particular the working group's experience.	p	req.	–	P	–	
G-4.2	The profile avoids a negative impact on existing implementations.	p	req.	–	U	–	
G-5 Assess any new terms carefully and avoid conflicting with ongoing standardisation. *(GeoDCAT-AP 1.0 ğ4.1)*							
G-5.1	The profile defines new terms only when existing vocabularies offer no suitable candidate, and only after careful assessment.	p	req.	–	P	–	
G-5.2	The profile discards new terms that could conflict with standards under preparation.	p	req.	–	N	–	
G-6 Cover the ISO core and common INSPIRE metadata elements; exclude only optional material where necessary, with theme-specific extensions outside the common scope. *(GeoDCAT-AP 1.0 ğ4.2)*							
G-6.1	The profile covers all metadata elements of the core profile of ISO 19115.	p	req.	–	S	–	
G-6.2	The profile covers the INSPIRE metadata elements, leaving out only those not common to all INSPIRE spatial data themes.	p	req.	–	P	–	
G-6.3	The profile excludes only optional elements.	p	req.	–	U	–	
G-7 Offer an additional RDF publication route while retaining the existing INSPIRE metadata obligations and implementations. *(GeoDCAT-AP 1.0 ğ1.1)*							
G-7.1	The profile offers an RDF syntax binding as an additional way to implement and publish INSPIRE/ISO metadata on DCAT-AP-conformant portals.	p	req.	–	U	–	
G-7.2	The profile leaves the INSPIRE metadata regulation and technical guidelines, and the collections maintained under them, unchanged.	p	req.	–	N	–	
G-8† Identify who is responsible for a metadata record and how to contact them. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.16, II.20)*							
G-8.1	Users can know who is responsible for a metadata record and how to contact them.	c	req.	extends	S	–	✓
G-9† Record the date of each metadata record. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.11, II.21)*							
G-9.1	Users can know the date of each metadata record.	c	req.	restates	S	–	✓
G-10† State the language of each metadata record. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.7, II.22)*							
G-10.1	Users can know the language of each metadata record.	c	req.	restates	S	–	✓
G-11† State the character set of each metadata record. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.19, II.24)*							
G-11.1	Users can know the character set of a metadata record.	c	req.	beyond	S	–	✓
G-12† Give each metadata record its own identifier. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.17)*							
G-12.1	Users can know the file identifier of a metadata record.	c	req.	extends	S	–	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
G-13†	State the metadata standard a record follows. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.18)*						
G-13.1	Users can know the metadata standard a record follows.	c	req.	restates	S	–	✓
G-14†	State the version of the metadata standard a record follows. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.18)*						
G-14.1	Users can know the version of the metadata standard a record follows.	c	req.	extends	S	–	✓
G-15†	Give each resource a title. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.1)*						
G-15.1	Users can know the title of each resource.	c	req.	restates	S	–	✓
G-16†	Give the dates of creation, publication and last revision of a resource. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.11)*						
G-16.1	Users can know the date of creation of a resource.	c	req.	beyond	S	–	✓
G-16.2	Users can know the date of publication of a resource.	c	req.	restates	S	–	✓
G-16.3	Users can know the date of last revision of a resource.	c	req.	restates	S	–	✓
G-17†	Describe each resource in a short abstract. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.2)*						
G-17.1	Users can know a short abstract describing each resource.	c	req.	restates	S	–	✓
G-18†	State the language or languages used in a resource. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.7)*						
G-18.1	Users can know the language or languages used in a resource.	c	req.	restates	S	–	✓
G-19†	Classify each resource by topic category. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.8)*						
G-19.1	Users can know the topic category of each resource.	c	req.	beyond	S	–	✓
G-20†	Give the geographic extent of a resource as a bounding box or a geographic identifier. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.10)*						
G-20.1	Users can know the geographic extent of a resource, as a bounding box or a geographic identifier.	c	req.	restates	S	–	✓
G-21†	State the character encoding of a dataset. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.24)*						
G-21.1	Users can know the character encoding of a dataset.	c	req.	beyond	S	–	✓
G-22†	Give the temporal extent a resource covers. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.11)*						
G-22.1	Users can know the temporal extent a resource covers.	c	req.	restates	S	–	✓
G-23†	Describe the lineage of a dataset: how it was produced and processed. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.12)*						
G-23.1	Users can know the lineage of a dataset: how it was produced and processed.	c	req.	restates	S	–	✓
G-24†	State how spatial data are represented, e.g. as vector or grid data. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.26)*						
G-24.1	Users can know how spatial data are represented, e.g. as vector or grid data.	c	req.	beyond	S	–	✓
G-25†	State the format in which each distribution is encoded. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.25)*						
G-25.1	Users can know the format in which each distribution is encoded.	c	req.	restates	S	–	✓
G-26†	State the spatial resolution of a dataset. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.13)*						
G-26.1	Users can know the spatial resolution of a dataset.	c	req.	beyond	S	–	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
G-27†	Identify the organisations responsible for a resource and the role each plays. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.16)*						
G-27.1	Users can know the organisations responsible for a resource and the role each plays.	c	req.	partly	S	–	✓
G-28†	Give links to the resource or to further information about it. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.4)*						
G-28.1	Users can access the resource, or further information about it, through links.	c	req.	restates	S	–	✓
G-29†	State the coordinate reference system of a dataset. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.23)*						
G-29.1	Users can know the coordinate reference system of a dataset.	c	req.	extends	S	–	✓
G-30†	State the temporal reference system of a dataset. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.23)*						
G-30.1	Users can know the temporal reference system of a dataset.	c	req.	extends	S	–	✓
G-31†	State the specifications a resource conforms to, and whether it conforms. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.14)*						
G-31.1	Users can know the specifications a resource conforms to, and whether it conforms.	c	req.	partly	S	–	✓
G-32†	State whether a resource is a dataset, a dataset series or a service. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.3)*						
G-32.1	Users can know whether a resource is a dataset, a dataset series or a service.	c	req.	partly	S	–	✓
G-33†	State the type of a spatial data service, e.g. discovery, view or download. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.9)*						
G-33.1	Users can know the type of a spatial data service, e.g. discovery, view or download.	c	req.	beyond	S	–	✓
G-34†	Characterise each resource with keywords and name the vocabularies they come from. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.8)*						
G-34.1	Users can know the keywords that characterise each resource and the vocabularies they come from.	c	req.	partly	S	–	✓
G-35†	Link a service to the datasets or dataset series it operates on. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.6)*						
G-35.1	Users can find the datasets or dataset series a service operates on.	c	req.	beyond	S	–	✓
G-36†	Give each resource a unique identifier. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.5)*						
G-36.1	Users can know the unique identifier of each resource.	c	req.	restates	S	–	✓
G-37†	State the conditions for accessing and using a resource. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.15)*						
G-37.1	Users can know the conditions for accessing and using a resource.	c	req.	restates	S	–	✓
G-38†	State any limitations on public access to a resource. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.15)*						
G-38.1	Users can know any limitations on public access to a resource.	c	req.	restates	S	–	✓
G-39†	State how often a resource is maintained and updated. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.27)*						
G-39.1	Users can know how often a resource is maintained and updated.	c	req.	partly	S	–	✓
G-40†	Report the topological consistency of a dataset as a data quality result. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.14; ğ5.3)*						
G-40.1	Users can know the topological consistency of a dataset, as a data quality result.	c	req.	beyond	S	–	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
G-41†	Report the conceptual and domain consistency of a dataset as data quality results. *(GeoDCAT-AP 1.0 ğ4.2; Annex I (Table 5); Annex II.14)*						
G-41.1	Users can know the conceptual consistency of a dataset, as a data quality result.	c	req.	beyond	S	–	✓
G-41.2	Users can know the domain consistency of a dataset, as a data quality result.	c	req.	beyond	S	–	✓
G-42†	Make spatial datasets, dataset series and services searchable on general data portals. *(GeoDCAT-AP 1.0 ğ1; ğ3)*						
G-42.1	The profile makes spatial datasets, dataset series and services searchable on general data portals.	p	req.	–	P	–	
G-43†	Allow ISO 19115/INSPIRE metadata records to be converted into the RDF representation; conversion back is not required. *(GeoDCAT-AP 1.0 ğ1.1)*						
G-43.1	The syntax binding supports the conversion of ISO 19115/INSPIRE metadata records into the RDF representation; conversion back is not required.	s	req.	–	U	–	
G-44†	Keep the RDF binding as simple as possible, reusing existing vocabularies and avoiding new terms. *(GeoDCAT-AP 1.0 ğ1.1)*						
G-44.1	The profile keeps its RDF syntax binding as simple as possible.	p	req.	–	U	–	
G-45†	Reuse the published alignments between EuroVoc, GEMET and the INSPIRE themes. *(GeoDCAT-AP 1.0 ğ1.1)*						
G-45.1	The profile reuses the published alignments between EuroVoc, GEMET and the INSPIRE themes.	p	req.	–	U	–	
G-46†	Feed recommendations into the DCAT-AP revision so that DCAT-AP and GeoDCAT-AP stay aligned. *(GeoDCAT-AP 1.0 ğ1.1)*						
G-46.1	The GeoDCAT-AP working group feeds recommendations into the DCAT-AP revision so that DCAT-AP and GeoDCAT-AP stay aligned.	s	req.	–	N	–	

HealthDCAT-AP HealthData@EU Pilot D6.2 (2024); base DCAT-AP 3.0.0; inspected during development: reference

69 rows (16 published), 121 atoms, 89 scored (51 requirements), 13 restate the base, 53 stated (85 at least in part); pre-consolidation corpus: 5 (22)

H-1	Use stable, resolvable identifiers so metadata records and their relationships remain reliably identifiable across catalogues. *(D6.2 ğ5.2.1.1)*						
H-1.1	Users can access each dataset's metadata record through a persistent, dereferenceable identifier (an HTTP URI).	c	req.	tightens	P	P	✓
H-1.2	Users can access the datasets and external resources a dataset links to through persistent, dereferenceable URIs.	c	req.	tightens	S	U	✓
H-2	Represent catalogue relationships as machine-interpretable links that support a health-data knowledge graph. *(D6.2 ğ5.2.1.2)*						
H-2.1	The profile represents the relationships between datasets and other resources (publications, codes) as machine-interpretable links, so the catalogue forms a knowledge graph.	p	req.	–	S	N	
H-3	Have data holders review their dataset descriptions at least once a year for accuracy and currency, and keep track of when each description was last reviewed. *(D6.2 ğ5.2.1.3)*						
H-3.1	Data holders check at least once a year that their dataset descriptions are accurate and up to date.	s	req.	–	P	N	
H-3.2	Users can know when a dataset description was last reviewed, separately from when it was last modified, where catalogues manage the review.	c	prop.	beyond	N	N	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-4 Make metadata sufficiently interpretable and informative to support machine-assisted search, including semantic and generative-AI use. *(D6.2 ğ5.2.2)*							
H-4.1	Users can find datasets with natural-language and AI-based search, through meaningful free-text descriptions (what, why, how, who).	c	req.	partly	P	S	u
H-4.2	Users can find datasets by querying their concepts and relationships semantically.	c	req.	beyond	S	N	✓
H-5 Allow dataset discovery to be refined using health categories, access bodies, holder types, personal-data categories, age ranges, population size, quality labels and coding information. *(D6.2 ğ5.2.3)*							
H-5.1	Users can filter datasets by health category in a category facet, using the controlled list of EHDS Art. 33 categories.	c	req.	beyond	P	P	✓
H-5.2	Users can filter datasets by health data access body.	c	req.	beyond	P	P	✓
H-5.3	Users can filter datasets by publisher type.	c	req.	extends	P	P	✓
H-5.4	Users can filter datasets by the categories of personal data they contain.	c	req.	beyond	P	P	✓
H-5.5	Users can filter population datasets by the minimum typical age of their population.	c	req.	beyond	S	P	✓
H-5.6	Users can filter population datasets by the maximum typical age of their population.	c	req.	beyond	S	P	✓
H-5.7	Users can filter datasets by population size.	c	req.	beyond	P	P	✓
H-5.8	Users can filter datasets by quality label.	c	req.	beyond	S	P	✓
H-5.9	Users can filter datasets by the coding values they use.	c	req.	beyond	S	P	✓
H-6 Adapt displayed filters to whether the catalogue query concerns personal or non-personal data. *(D6.2 ğ5.2.3)*							
H-6.1	The catalogue adapts the displayed search filters to whether the user is looking for personal or non-personal data.	s	req.	–	U	N	
H-7 Use semantic annotations and shared coding vocabularies to make health descriptions interoperable and searchable. *(D6.2 ğ5.2.5)*							
H-7.1	Users can find the datasets about the same concept across the data space, through shared concepts with persistent identifiers.	c	req.	beyond	S	U	✓
H-7.2	Users can know the health themes of a dataset, as HTTP URIs from an authoritative vocabulary.	c	req.	beyond	P	N	✓
H-7.3	Users can know the coding systems a dataset uses, as HTTP URIs.	c	req.	beyond	S	N	✓
H-7.4	Users can know the code values a dataset uses, linked to external standards such as ICD-10.	c	req.	beyond	S	N	✓
H-8 Describe at least one safe sample distribution for non-public datasets so users can prepare analyses without accessing sensitive records. *(D6.2 ğ5.2.6)*							
H-8.1	Users can access at least one sample distribution (mock-up, synthetic or anonymised data) of every non-public dataset.	c	req.	tightens	P	S	✓
H-9 Accompany sample data with definitions of its variables in a data dictionary. *(D6.2 ğ5.2.6)*							
H-9.1	Users can access a data dictionary that defines all data elements of each sample distribution.	c	req.	beyond	P	N	✓
H-10 Allow applicants to identify the variables they need, supporting data minimisation. *(D6.2 ğ5.2.6 (pp. 43–44))*							
H-10.1	Data access applicants can list the specific variables they need, in line with data minimisation.	s	req.	–	S	N	
H-11 Provide usable descriptions of data structures and coding so datasets can be understood and processed consistently. *(D6.2 ğ5.2.6.1)*							
H-11.1	Users can know the data structures of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-11.2	Users can know the data formats of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.3	Users can know the vocabularies of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.4	Users can know the classification schemes of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-11.5	Users can know the taxonomies of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	P	N	✓
H-11.6	Users can know the code lists of a dataset, described in a publicly available and consistent manner, where available.	c	req.	beyond	S	N	✓
H-12 Describe logical dataset subdivisions and their relationships where large holdings need to be partitioned. *(D6.2 ğ5.2.6.2)*							
H-12.1	Users can know the coherent logical ('virtual') datasets within a large data warehouse, each with its own metadata record.	c	req.	restates	P	U	✓
H-12.2	Users can know how a data source breaks down into collections and datasets, from the dataset descriptions.	c	def.	?	P	U	
H-13 State intellectual-property and reuse conditions, including those on sample distributions. *(D6.2 ğ5.2.6.3)*							
H-13.1	Users can know the conditions for re-use (rights) of every distribution, including sample distributions.	c	req.	tightens	P	N	✓
H-13.2	Users can know which parts of a dataset are protected by intellectual property rights.	c	req.	extends	P	N	✓
H-13.3	Users can know why parts of a dataset need intellectual-property protection.	c	req.	extends	N	N	✓
H-13.4	Users can access a redacted data dictionary, when intellectual property rights prevent a sample subset.	c	prop.	beyond	N	N	✓
H-14 Link to descriptive analytics that help prospective users assess a dataset. *(D6.2 ğ5.2.7)*							
H-14.1	Users can access analytics resources (dashboards, reports, statistics APIs) to assess a dataset without accessing the raw data.	c	req.	beyond	P	S	✓
H-15 Expose quality annotations or labels that help assess fitness for reuse. *(D6.2 ğ5.2.8)*							
H-15.1	Users can know the quality or utility annotations (e.g. a quality label or certificate) of a dataset or distribution.	c	req.	beyond	S	N	✓
H-15.2	Users can know the motivation of each quality annotation.	c	req.	beyond	P	N	✓
H-15.3	Users can know the content of each quality annotation, as text or as a link to a resource.	c	req.	beyond	P	N	✓
H-15.4	The profile describes dataset quality with one standardised approach across EU data spaces, aligned with StatDCAT-AP and mobilityDCAT-AP (DQV quality annotations).	p	req.	–	P	N	
H-16 Describe the purpose of data collection and the legal basis for processing personal data. *(D6.2 ğ5.2.9)*							
H-16.1	Users can know the purpose for which the data of a dataset were collected.	c	req.	beyond	S	S	✓
H-16.2	Users can know the legal basis for processing the personal data in a dataset.	c	req.	beyond	S	N	✓
H-17† Describe each dataset's source, scope, main characteristics, the nature of its health data and the conditions for making the data available. *(D6.2 ğ4 (quoting EHDS Art. 55))*							
H-17.1	Users can know the source of each dataset.	c	req.	tightens	S	P	✓
H-17.2	Users can know the scope of each dataset.	c	req.	tightens	S	U	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-17.3	Users can know the main characteristics of each dataset.	c	req.	restates	S	U	✓
H-17.4	Users can know the nature of the electronic health data in each dataset.	c	req.	beyond	S	N	✓
H-17.5	Users can know the conditions for making the data of a dataset available.	c	req.	tightens	S	N	✓
H-18†	Indicate the legislation that applies to a dataset (e.g. the EHDS Regulation), so health catalogues can filter the datasets in its scope. *(D6.2 ğ4.2; ğ5.3.1.1; ğ5.3.1.3 (Applicable legislations))*						
H-18.1	Users can know the legislation that applies to each dataset (e.g. the EHDS Regulation), so catalogues can filter the datasets in its scope.	c	prop.	tightens	P	N	✓
H-19†	Classify each dataset's access rights as public or non-public, so catalogues know whether access needs an application. *(D6.2 ğ5.3.1.3 (Access rights))*						
H-19.1	Users can know each dataset's access rights (public, restricted or non-public), so catalogues know whether access needs an application; for NSIP data the rights are only 'non-public' or 'restricted'.	c	prop.	tightens	P	U	✓
H-20†	Provide at least one distribution for every dataset: an access URL or download link for public data, a landing page of the health data access body for non-public data. *(D6.2 ğ5.3.1.1; ğ5.3.1.3 (Access rights; Dataset distribution))*						
H-20.1	Users can access at least one distribution of every dataset.	c	prop.	tightens	P	N	✓
H-20.2	Users can access a public dataset through an access URL or download link, and a non-public dataset through the landing page of the health data access body or equivalent authority.	c	prop.	tightens	P	N	✓
H-21†	Name a contact point for every dataset, so users can ask about it without working out whom to contact. *(D6.2 ğ5.3.1.3 (Contact point))*						
H-21.1	Users can know the contact point of every dataset, which acts as its helpdesk.	c	prop.	tightens	P	N	✓
H-22†	Identify the health data access body that acts as the access gateway for a non-public or personal dataset. *(D6.2 ğ5.3.1.3 (Health data access body))*						
H-22.1	Users can know the health data access body that acts as access gateway for every dataset with personal or non-public data.	c	prop.	beyond	P	N	✓
H-23†	Indicate whether a dataset contains personal data. *(D6.2 ğ5.3.1.3 (Data type); ğ5.3.1.4)*						
H-23.1	Users can know whether a dataset contains personal or non-personal data (data type).	c	prop.	extends	P	U	✓
H-24†	Give each dataset's geographical coverage, preferably from controlled vocabularies, as a filter for populations. *(D6.2 ğ5.3.1.3 (Geographical coverage))*						
H-24.1	Users can filter datasets by their geographical coverage, when considering populations.	c	prop.	tightens	S	P	✓
H-25†	Characterise each dataset with a few main keywords per language rather than long lists. *(D6.2 ğ5.3.1.3 (Keyword))*						
H-25.1	Users can find each dataset by a few main keywords per language (10–15) rather than long lists.	c	prop.	tightens	P	N	✓
H-26†	Describe a dataset's lineage: how it was created and modified, and by whom. *(D6.2 ğ5.3.1.3 (Provenance))*						
H-26.1	Users can know a dataset's lineage: how it was created and modified, and by whom.	c	prop.	tightens	S	S	✓
H-27†	Identify the publisher as the data holder accountable for making the dataset accessible. *(D6.2 ğ5.3.1.3 (Publisher))*						
H-27.1	Users can know the publisher of a dataset, as the data holder accountable for making it accessible.	c	prop.	tightens	P	N	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-28†	Describe the nature and type of the data holder as context for judging a dataset's credibility. *(D6.2 ğ5.3.1.3 (Publisher note; Publisher type))*						
H-28.1	Users can know the nature of the data holder, as context for judging the credibility of the dataset.	c	prop.	beyond	P	N	✓
H-28.2	Users can know the publisher type of the data holder, from a controlled vocabulary.	c	prop.	tightens	S	N	✓
H-29†	Tag every dataset in a health catalogue with the EU data theme for health. *(D6.2 ğ5.3.1.3 (Theme); ğ5.3.1.4)*						
H-29.1	Users can find every dataset in a health catalogue by the EU data theme for health.	c	prop.	tightens	P	N	✓
H-30†	Identify the data models and other standards a dataset conforms to (e.g. OMOP), so users can find data ready for their analysis. *(D6.2 ğ4.1; ğ5.2.6.2.1; ğ5.3.1.3 (Conforms to))*						
H-30.1	Users can know the data model or other standard a dataset conforms to (e.g. OMOP), so they can find data ready for their analysis.	c	req.	tightens	S	N	✓
H-30.2	Users can use machine-interpretable identifiers (HTTP URIs) for the standards a dataset conforms to.	c	req.	tightens	S	N	✓
H-31†	Link to documentation that helps users understand the dataset. *(D6.2 ğ5.3.1.3 (Documentation))*						
H-31.1	Users can access documentation that helps them understand the dataset.	c	prop.	tightens	S	N	✓
H-32†	Link a dataset to related resources and publications that show how it has been used. *(D6.2 ğ5.2.1.2; ğ5.3.1.3 (Is referenced by; Related resource); ğ6.1)*						
H-32.1	Users can find the publications and other resources that reference a dataset.	c	prop.	tightens	S	N	✓
H-32.2	Users can find the resources related to a dataset.	c	prop.	tightens	S	N	✓
H-33†	Identify the datasets a dataset was derived from, and let datasets derived from it refer back to it. *(D6.2 ğ5.3.1.3 (Source); ğ6.1)*						
H-33.1	Users can know the datasets a dataset was derived from; research datasets derived from a health dataset cite its metadata identifier.	c	prop.	tightens	P	U	✓
H-34†	Give a landing page that provides access to the dataset and further information. *(D6.2 ğ5.3.1.3 (Landing page))*						
H-34.1	Users can access the dataset and further information through a landing page.	c	prop.	tightens	S	N	✓
H-35†	State the language of a dataset where applicable. *(D6.2 ğ5.3.1.3 (Language))*						
H-35.1	Users can know the language of a dataset, where applicable.	c	prop.	tightens	S	N	✓
H-36†	Describe the population a dataset covers, where it contains one. *(D6.2 ğ5.3.1.3 (Population coverage))*						
H-36.1	Users can know the population a dataset covers, where it contains one.	c	prop.	beyond	S	S	✓
H-37†	State the size of a dataset (number of records, number of unique individuals; byte size of each distribution), so users can judge its value. *(D6.2 ğ2.3.1.3 and ğ5.3.1.2 (DGA Art. 8); ğ5.3.1.3 (Number of records))*						
H-37.1	Users can know the number of records in a dataset.	c	prop.	beyond	S	P	✓
H-37.2	Users can know the number of unique individuals in a dataset.	c	prop.	beyond	P	P	✓
H-37.3	Users can know the byte size of each distribution.	c	prop.	tightens	P	U	✓
H-38†	Give a dataset's temporal coverage, leaving the end open while collection continues. *(D6.2 ğ5.3.1.3 (Temporal coverage))*						
H-38.1	Users can know a dataset's temporal coverage, with the end left open while collection continues.	c	prop.	restates	S	N	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-39†	State a dataset's temporal resolution, to judge whether it can be reused. *(D6.2 ğ5.3.1.3 (Temporal resolution))*						
H-39.1	Users can know a dataset's temporal resolution, to judge whether it can be reused.	c	prop.	tightens	S	N	✓
H-40†	State when a dataset must be deleted (retention period), and keep its record available afterwards. *(D6.2 ğ5.3.1.3 (Retention period))*						
H-40.1	Users can know the retention period after which a dataset is to be deleted.	c	prop.	beyond	P	N	✓
H-40.2	The catalogue keeps a deleted dataset's metadata record available, because publications can refer to it.	s	prop.	–	U	U	
H-41†	Give a version designation where the holder follows a versioning strategy. *(D6.2 ğ5.3.1.3 (Version))*						
H-41.1	Users can know the version designation of a dataset, where the holder follows a versioning strategy.	c	prop.	restates	S	N	✓
H-42†	State the spatial resolution where it applies, mainly for geospatial data. *(D6.2 ğ5.3.1.3 (Spatial resolution))*						
H-42.1	Users can know the spatial resolution of a dataset where it applies, mainly for geospatial data.	c	prop.	restates	S	N	✓
H-43†	Record when each catalogue entry was listed and last changed, and which original metadata it was created from. *(D6.2 ğ5.2.1.1)*						
H-43.1	Users can know when each catalogue entry was listed.	c	req.	restates	S	N	✓
H-43.2	Users can know when each catalogue entry was last modified.	c	req.	restates	S	N	✓
H-43.3	Users can know the original metadata an aggregated catalogue entry was created from.	c	req.	restates	S	U	✓
H-44†	State which application profile a metadata record follows, so aggregators can validate it. *(D6.2 ğ5.2.1.3)*						
H-44.1	Users can know which application profile a metadata record follows, so aggregators can validate it.	c	req.	restates	S	U	✓
H-45†	Indicate whether a dataset's holder is a trusted data holder, as a search facet. *(D6.2 ğ5.2.3 (facet list))*						
H-45.1	Users can filter datasets by whether their holder is a trusted data holder, in a search facet.	c	prop.	beyond	N	U	✓
H-46†	Provide the information the EHDS data quality and utility label needs: documentation, maturity of quality management, coverage, access and provision times, and data modifications. *(D6.2 ğ5.2.8 (quoting EHDS Art. 56))*						
H-46.1	Users can know, for the EHDS data quality and utility label, how representative the sampled population is, when applicable.	c	req.	beyond	S	P	✓
H-46.5	Users can know, for the EHDS data quality and utility label, the average time a natural person appears in the dataset.	c	req.	beyond	S	P	✓
H-46.2	Users can know, for the EHDS data quality and utility label, the maturity of the data quality management processes, including review and audit processes and the examination of biases.	c	req.	beyond	S	N	✓
H-46.3	Users can know, for the EHDS data quality and utility label, the time between collecting the data and adding them to the dataset.	c	req.	beyond	S	N	✓
H-46.6	Users can know, for the EHDS data quality and utility label, the time to provide the data after a data access application is approved.	c	req.	beyond	S	N	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-46.4	Users can know, for the EHDS data quality and utility label, the modifications of the data: merging and adding data to an existing dataset, including links with other datasets.	c	req.	beyond	S	P	✓
H-47† Standardise the minimum metadata elements that health data holders must provide under the EHDS Regulation. *(D6.2 ğ2.2)*							
H-47.1	The profile standardises the minimum metadata elements that health data holders are required to provide under the EHDS Regulation.	p	req.	–	S	N	
H-48† Remain fully compatible with DCAT-AP. *(D6.2 ğ2.2)*							
H-48.1	The profile remains fully compatible with DCAT-AP.	p	req.	–	S	N	
H-49† Align the minimum elements for open health data with those of DCAT-AP for High-Value Datasets. *(D6.2 ğ4.2; ğ5.3.1.1)*							
H-49.1	The profile aligns the minimum elements for open health data with those of DCAT-AP for High-Value Datasets.	p	prop.	–	P	N	
H-50† Align the minimum elements for non-open, non-personal health data with the metadata of national single information points under the Data Governance Act. *(D6.2 ğ5.3.1.2)*							
H-50.1	The profile aligns the minimum elements for non-open, non-personal health data with the NSIP metadata of the Data Governance Act.	p	prop.	–	P	N	
H-51† Define stricter, conditional minimum element sets for protected and for sensitive (personal) health data. *(D6.2 ğ4.2; ğ5.3.1.4)*							
H-51.1	The profile defines stricter, conditional minimum element sets for protected and for sensitive (personal) health data.	p	req.	–	P	N	
H-52† Take coded values (health themes, coding systems, code values, standards) as HTTP URIs from one shared, authoritative knowledge base (Wikidata). *(D6.2 ğ5.2.4; ğ5.2.5)*							
H-52.1	The profile takes coded values (health themes, coding systems, code values, standards) as HTTP URIs from one shared, authoritative knowledge base (Wikidata).	p	req.	–	P	U	
H-53† Publish data dictionaries in one harmonised machine-readable format (CSVW), so variable definitions are consistent across catalogues. *(D6.2 ğ5.2.6.1)*							
H-53.1	The profile provides for data dictionaries in one harmonised machine-readable format (CSVW).	p	req.	–	P	N	
H-54† Provide descriptions in at least one official EU language, preferably the holder's native language, with language tags. *(D6.2 ğ6.1.3 (quoting EHDS Art. 55(1a)))*							
H-54.1	The profile provides for dataset descriptions in at least one official EU language.	p	req.	–	P	N	
H-54.2	The profile encourages data holders to create metadata in their native language.	p	req.	–	P	N	
H-55† Expose national health catalogues through endpoints that the EU datasets catalogue, single information points and other catalogues can harvest. *(D6.2 ğ2.3.1.3; ğ6.2)*							
H-55.1	Health data access bodies expose the national health catalogues through endpoints that the EU datasets catalogue, single information points and other catalogues can harvest.	s	req.	–	S	P	
H-56† Keep the data holder's record as the single source of truth when aggregating: copies refer to the original, and access bodies add a persistent identifier without replacing the original one. *(D6.2 ğ5.2.1.1)*							
H-56.1	Aggregating catalogues keep their copies of metadata linked to the original record at its source.	s	req.	–	S	P	
H-56.2	Data holders update their metadata only at its primary source; they, not health data access bodies, own and edit it.	s	req.	–	P	U	

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-56.3	Health data access bodies assign an additional persistent URI to a record whose identifier does not meet the profile's identifier rules, keeping the data holder's original identifier.	s	req.	–	P	P	
H-57†	Detect broken URIs in the links between catalogue resources. *(D6.2 ğ5.2.1.2)*						
H-57.1	Those operating the health data space can detect broken URIs in the links between catalogue resources.	s	req.	–	U	N	
H-58†	Measure the maturity of the health data space from its metadata, such as the share of URIs, the level of interoperability and conformity to standards. *(D6.2 ğ5.2.1.2)*						
H-58.1	Those governing the health data space measure its maturity from its metadata (share of URIs, interoperability, conformity to standards).	s	req.	–	P	N	
H-59†	Order search facets by their relevance, as shown in the pilot use cases. *(D6.2 ğ5.2.3)*						
H-59.1	The search interface orders facets by their relevance, as shown in the pilot use cases.	s	req.	–	N	U	
H-60†	Let applicants attach an analysis script to a data request, to obtain anonymised statistical answers. *(D6.2 ğ5.2.6 (EHDS Art. 47))*						
H-60.1	Applicants can attach an analysis script to a data request, to obtain anonymised statistical answers.	s	req.	–	P	U	
H-61†	Support data holders in transforming existing metadata into the profile, with a user-friendly multilingual editor that needs no technical expertise. *(D6.2 ğ6; ğ6.1)*						
H-61.1	Data holders get support in transforming their existing metadata into the profile.	s	req.	–	S	P	
H-61.2	Data holders get a user-friendly, multilingual metadata editor that needs no technical expertise.	s	req.	–	P	P	
H-62†	Validate metadata against the profile (mandatory elements, data types, controlled vocabularies) before publication and when aggregating. *(D6.2 ğ6.1.2; ğ6.2.1)*						
H-62.1	Data holders and metadata aggregators validate metadata against the profile (mandatory elements, data types, controlled vocabularies): data holders before publication, aggregators when aggregating.	s	req.	–	S	U	
H-63†	Assign responsible metadata managers at data holders and access bodies; the data holder, not the access body, owns and edits the metadata. *(D6.2 ğ6.1)*						
H-63.1	Data holders and health data access bodies each designate a responsible metadata manager.	s	req.	–	P	N	
H-64†	Let access bodies enrich metadata automatically when indicators show that a dataset falls under the EHDS. *(D6.2 ğ5.3.1.1)*						
H-64.1	Health data access bodies enrich metadata automatically (e.g. add the EHDS as applicable legislation) when indicators show a dataset falls under the EHDS.	s	prop.	–	U	U	
H-65†	Set up revision and maintenance procedures, with clear responsibility, for the profile. *(D6.2 ğ6.2)*						
H-65.1	The profile's maintainers set up revision and maintenance procedures, with clear responsibility.	s	prop.	–	U	N	
H-66†	Provide the elements DCAT-AP already makes mandatory: a title and a description for every dataset and an access URL for every distribution. *(D6.2 ğ5.3.1.1; ğ5.3.1.2 (NSIP table); ğ5.3.1.3 (Title; Description))*						
H-66.1	Users can know the title of every dataset.	c	prop.	restates	S	N	✓
H-66.2	Users can know the description of every dataset.	c	prop.	restates	S	N	✓
H-66.3	Users can access the data of each distribution, or information on how to request it, through an access URL.	c	prop.	restates	S	N	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
H-67† Give each distribution of protected (NSIP) data exactly one file format, and offer other formats as separate distributions. *(D6.2 ğ5.3.1.2 (NSIP distribution table))*							
H-67.1	Users can know the file format of each distribution, with exactly one format per distribution for protected (NSIP) data.	c	prop.	tightens	P	N	✓
H-67.2	Users can access the same data in another format as a separate distribution.	c	prop.	restates	S	N	✓
H-69† Offer a common harmonised data validator in the infrastructure to check datasets that follow common data models. *(D6.2 ğ5.2.6.2.1)*							
H-69.1	The infrastructure offers a common harmonised data validator that checks the quality and consistency of datasets that follow common data models.	s	prop.	–	P	N	
H-68† State how often a dataset is updated (frequency). *(D6.2 ğ5.3.1.3 (recommended properties: frequency))*							
H-68.1	Users can know how often a dataset is updated (frequency).	c	prop.	tightens	S	N	✓

StatDCAT-AP StatDCAT-AP 1.0.0 (2016); base DCAT-AP 1.1; inspected during development: reference

22 rows (7 published), 27 atoms, 14 scored (10 requirements), 6 restate the base, 4 stated (10 at least in part)

S-1 Expose the dimensions that distinguish observations, including time and demographic or regional dimensions. *(StatDCAT-AP 1.0 ğ7.2.1)*							
S-1.1	Users can know the dimensions that identify observations in a statistical dataset (e.g. time, sex, age, region), described the same way whatever the distribution format.	c	req.	beyond	P	–	✓
S-2 Expose attributes needed to interpret observations, such as scaling and observation status. *(StatDCAT-AP 1.0 ğ7.2.1)*							
S-2.1	Users can know the attributes needed to qualify and interpret observed values (e.g. units of measure, scaling factors, observation status), described the same way whatever the distribution format.	c	req.	beyond	P	–	✓
S-3 Associate statistical datasets with descriptive quality information. *(StatDCAT-AP 1.0 ğ7.2.2)*							
S-3.1	Users can know the quality information of a dataset or distribution.	c	req.	beyond	N	–	✓
S-3.2	Users can know the motivation of each quality annotation.	c	req.	beyond	N	–	✓
S-3.3	Users can know the quality information, as text or as a link to quality information published elsewhere.	c	req.	beyond	N	–	✓
S-4 Identify delivery resources that present dataset visualisations. *(StatDCAT-AP 1.0 ğ7.2.3)*							
S-4.1	Users can access visualisations of the data (tables, graphs or interactive views).	c	req.	extends	P	–	✓
S-5 Indicate the actual number of statistical series represented in a dataset. *(StatDCAT-AP 1.0 ğ7.2.4)*							
S-5.1	Users can know the actual number of data series in a dataset, as distinct from the number of observations and the theoretical number of combinations.	c	req.	beyond	N	–	✓
S-6 Identify the unit in which dataset observations are measured. *(StatDCAT-AP 1.0 ğ7.2.5)*							
S-6.1	Users can know the unit of measurement of the observations in a dataset.	c	req.	beyond	S	–	✓
S-7 Represent time-series coverage and periodicity without relying on an ambiguous shorthand. *(StatDCAT-AP 1.0 ğ7.2.6)*							
S-7.1	Users can know the start of the period a time series covers.	c	guid.	restates	S	–	✓
S-7.2	Users can know the end of the period a time series covers.	c	guid.	restates	S	–	✓
S-7.3	Users can know the periodicity of a time series.	c	guid.	restates	P	–	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
S-8†	Improve the discoverability of statistical datasets on general open data portals. *(StatDCAT-AP 1.0.0 ğ2.2 (charter); ğ5.1)*						
S-8.1	The profile improves the discoverability of statistical datasets on general open data portals.	p	req.	–	S	–	
S-9†	Let statistical portals feed general open data portals by exporting their metadata in the profile, with guidance for SDMX-based systems. *(StatDCAT-AP 1.0.0 ğ5.2; ğ8; Annexes IV–V)*						
S-9.1	Statistical portals can export their metadata in the profile, so general open data portals can take it up.	s	req.	–	S	–	
S-9.2	Organisations with SDMX-based systems get guidance for mapping and extracting their metadata into the profile.	s	req.	–	U	–	
S-10†	Conform fully to DCAT-AP 1.1, meeting every obligation of its conformance statement. *(StatDCAT-AP 1.0.0 ğ7.1)*						
S-10.1	The profile conforms fully to DCAT-AP and keeps statistical descriptions aligned with DCAT and DCAT-AP.	p	req.	–	P	–	
S-11†	Concentrate on metadata elements that help users discover datasets. *(StatDCAT-AP 1.0.0 ğ6.2)*						
S-11.1	The profile concentrates on the metadata elements that help users discover datasets.	p	req.	–	P	–	
S-12†	Take terms from existing, well-known and well-maintained vocabularies, including the ISA Core Vocabularies and Eurostat's metadata vocabularies. *(StatDCAT-AP 1.0.0 ğ6.2)*						
S-12.1	The profile takes its terms from existing, well-known and well-maintained vocabularies.	p	req.	–	S	–	
S-13†	Prefer common controlled vocabularies maintained by the EU Publications Office. *(StatDCAT-AP 1.0.0 ğ6.2)*						
S-13.1	The profile prefers common controlled vocabularies, preferably those maintained by the EU Publications Office.	p	req.	–	P	–	
S-14†	Balance simplicity and expressiveness for a wide, non-specialist audience. *(StatDCAT-AP 1.0.0 ğ6.2)*						
S-14.1	The profile balances simplicity and expressiveness for a wide, non-specialist audience.	p	req.	–	N	–	
S-15†	Make the profile usable for datasets in any format, such as SDMX, Data Cube or CSV. *(StatDCAT-AP 1.0.0 ğ7.1; Annex III)*						
S-15.1	The profile is usable for datasets in any format, such as SDMX, Data Cube or CSV.	p	req.	–	S	–	
S-16†	Classify statistical datasets by EU data theme, mapping Eurostat's own themes onto it. *(StatDCAT-AP 1.0.0 ğ7.8.3)*						
S-16.1	Users can find statistical datasets by EU data theme.	c	req.	restates	P	–	✓
S-17†	Link an RDF distribution to the data structure definition it follows. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-17.1	Users can know which data structure definition an RDF distribution follows.	c	guid.	restates	S	–	✓
S-18†	Record publication and modification times to the time of day when data are published more than once a day. *(StatDCAT-AP 1.0.0 ğ11; Annex III)*						
S-18.1	Users can know the publication and modification times of a dataset to the time of day, when its data are published more than once a day.	c	req.	restates	P	–	✓
S-19†	Identify the measures a statistical dataset contains, as SDMX and Data Cube define them. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-19.1	Users can know the measures a statistical dataset contains, as SDMX and Data Cube define them.	c	def.	beyond	S	–	

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
S-20†	Describe the quality of statistical data in more detail, e.g. following the ESS Single Integrated Metadata Structure (SIMS). *(StatDCAT-AP 1.0.0 §7.2.2; Annex III)*						
S-20.1	Users can know the quality of statistical data in more detail.	c	def.	beyond	N	–	
S-21†	Show how DDI descriptions of datasets can be expressed with the profile. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-21.1	The profile shows how DDI descriptions of datasets can be expressed with it.	p	def.	–	U	–	
S-22†	Document, with a full example, how datasets, distributions, Data Cube datasets and data structure definitions relate. *(StatDCAT-AP 1.0.0 Annex III; Annex VII)*						
S-22.1	The profile documents, with a full example, how datasets, distributions, Data Cube datasets and data structure definitions relate.	p	req.	–	P	–	

DCAT-AP-JRC Perego et al., SDSVoc 2016, paper 27; base DCAT-AP 1.1; inspected during development: reference

11 rows (5 published), 18 atoms, 17 scored (9 requirements), 4 restate the base, 4 stated (16 at least in part)

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
JRC-1	Identify the people who created a dataset. *(Metadata elements relevant for scientific data, numbered item 1)*						
JRC-1.1	Users can know the authors of a dataset.	c	req.	beyond	S	–	✓
JRC-2	Describe the origin and processing history of data. *(Metadata elements relevant for scientific data, numbered item 2)*						
JRC-2.1	Users can know the lineage of a dataset.	c	req.	restates	S	–	✓
JRC-3	Give prospective users practical guidance on using a dataset. *(Metadata elements relevant for scientific data, numbered item 3)*						
JRC-3.1	Users can know how to use the data of a dataset (usage notes).	c	req.	beyond	P	–	✓
JRC-4	Connect datasets to associated scientific publications. *(Metadata elements relevant for scientific data, numbered item 4)*						
JRC-4.1	Users can find the scientific publications about a dataset.	c	req.	beyond	P	–	✓
JRC-5	Identify source datasets used in producing a dataset. *(Metadata elements relevant for scientific data, numbered item 5)*						
JRC-5.1	Users can find the input data of a dataset (the data used to create it).	c	req.	restates	P	–	✓
JRC-6†	Give datasets persistent identifiers, encoded as HTTP URIs where possible, and tell primary identifiers (e.g. DOIs) from alternative ones. *(Perego et al. 2016 (SDSVoc paper 27), "Data citation", "Identifiers")*						
JRC-6.1	Users can reliably filter and link datasets by their authors and input data, through persistent identifiers.	c	req.	tightens	P	–	✓
JRC-6.2	Users can use identifiers encoded as HTTP URIs, wherever possible.	c	open	tightens	P	–	✓
JRC-6.3	Users can know which of a dataset's identifiers is the primary one (e.g. a DOI) and which are alternative ones.	c	req.	restates	S	–	✓
JRC-6.4	Users can cite a dataset by its DOI, when DataCite citations are generated.	c	req.	tightens	S	–	✓
JRC-6.5	Users can know the scheme or type of each identifier.	c	req.	restates	P	–	✓
JRC-7†	Express the roles that agents play for a dataset beyond publisher and contact point, consistently across catalogues. *(Perego et al. 2016 (SDSVoc paper 27), "Data citation", "Agent roles")*						
JRC-7.1	Users can know the roles that agents play for a dataset beyond publisher and contact point, from a minimal, consistent set of unambiguous roles.	c	open	beyond	P	–	✓

ID	Reference requirement (atom)	Type	Status	Base	Raw	Cons.	Sc.
JRC-8†	Indicate when a distribution gives access through a service or API rather than to the data itself, and describe the service interface with the query parameters needed to select data. *(Perego et al. 2016 (SDSVoc paper 27), "Modelling service/API-based data access")*						
JRC-8.1	Users can know when a distribution points to a service or API rather than directly to the data.	c	open	extends	P	–	✓
JRC-8.2	Users can know the API or service interface and the query parameters needed to select data, so software agents can use it directly.	c	open	beyond	P	–	✓
JRC-9†	Describe provenance in enough detail to support reproducibility and to track how data are used. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (detailed specification of data provenance))*						
JRC-9.1	Users can know the provenance of a dataset in enough detail to support reproducibility and to track how the data are used.	c	open	beyond	P	–	✓
JRC-10†	Describe data and metadata quality assessments and include users' feedback on the data. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (modelling and using data quality assessments))*						
JRC-10.1	Users can know the data quality assessments of a dataset.	c	open	beyond	P	–	✓
JRC-10.2	Users can know the feedback users have given on the data, as quality information.	c	open	beyond	P	–	✓
JRC-10.3	Users can know the quality assessments of the metadata of a dataset.	c	open	beyond	N	–	✓
JRC-11†	Publish metadata on the Web so that the data are visible and discoverable through search engines. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (optimising metadata publication on the Web))*						
JRC-11.1	The catalogue publishes its metadata on the Web, so that the data are visible and discoverable through search engines.	s	open	–	P	–	